

Descriptive Statistics

Session 2 — Cleaned-Up Study Notes

This guide reorganizes the original handwritten session notes into a clear, structured reference. Topics covered: quantiles & percentiles, the five-number summary, boxplots & outliers, scatterplots, covariance, correlation, and visualizing multiple variables — each with plain-language explanations and worked examples.

Contents

- 1. Quantiles & Percentiles
- 2. The Five-Number Summary
- 3. Boxplots & the Interquartile Range (IQR)
- 4. Scatterplots
- 5. Covariance
- 6. Correlation
- 7. Correlation vs. Causation
- 8. Visualizing Multiple Variables

1. Quantiles & Percentiles

A **quantile** is a value that splits sorted data into equal-sized groups. Quantiles help you understand how data is distributed, compare datasets, and spot outliers. Important: quantiles are **positions/locations** in sorted data, not new values you invent — data must always be sorted low to high first.

Type	Splits data into	Named points
Quartiles	4 equal parts	Q1 (25th pct), Q2 (50th pct / median), Q3 (75th pct)
Quintiles	5 equal parts	20th, 40th, 60th, 80th percentiles
Deciles	10 equal parts	D1 (10th) ... D9 (90th)
Percentiles	100 equal parts	P1 (1st) ... P99 (99th)

Key point: all other quantile types can be derived from percentiles (e.g., Q1 = P25, the median = P50, Q3 = P75).

What is a percentile?

The **p-th percentile** is the value below which p% of the observations fall. For example, the 75th percentile is the value below which 75% of the data lies.

Formula — finding the location of a percentile

$$\text{Location (L)} = (p / 100) \times N$$

where **p** = desired percentile rank, and **N** = total number of observations. This gives you the *position* in the sorted list — not the value itself. If L lands between two positions, you interpolate between them.

Formula — finding the percentile rank of a given value

$$\text{Percentile rank} = (X + 0.5 \cdot Y) / N \times 100$$

where **X** = number of values strictly below the given value, **Y** = number of values equal to the given value, and **N** = total number of values.

Worked Example: Find the 75th percentile

Dataset (already sorted): 78, 82, 84, 88, 91, 93, 94, 96, 98, 99 (N = 10)

Step 1 — Location: $L = (75/100) \times 10 = 7.5$

Step 2 — Since 7.5 falls between the 7th value (94) and 8th value (96), interpolate:

$$94 + 0.5 \times (96 - 94) = \mathbf{95}$$

So the 75th percentile of this dataset is approximately **95**.

2. The Five-Number Summary

The five-number summary condenses a dataset into five values that split it into four equal quarters (quartiles). It's a quick way to describe center, spread, and shape.

#	Value	Meaning
1	Minimum	The smallest value in the dataset
2	Q1 (1st quartile)	Separates the lowest 25% of data from the rest
3	Q2 (Median)	Separates the lowest 50% from the highest 50%
4	Q3 (3rd quartile)	Separates the lowest 75% from the highest 25%
5	Maximum	The largest value in the dataset

The five-number summary is usually displayed visually as a **boxplot** (see next section). It's a fast way to see the central tendency, variability, and overall distribution shape of a dataset.

Interquartile Range (IQR)

$$\text{IQR} = Q3 - Q1$$

The IQR measures the spread of the *middle 50%* of the data. It's a robust measure of variability — much less affected by extreme values than the full range.

3. Boxplots & the Interquartile Range

What is a boxplot?

A boxplot (box-and-whisker plot) is a graphical view of the five-number summary. The 'box' spans from Q1 to Q3 (the IQR), with a line at the median (Q2). 'Whiskers' extend out to the minimum and maximum non-outlier values, and any points beyond that are plotted individually as outliers.

Outlier fences

$$\text{Lower fence} = Q1 - 1.5 \times \text{IQR} \quad | \quad \text{Upper fence} = Q3 + 1.5 \times \text{IQR}$$

Any data point falling outside these fences is flagged as a potential **outlier**.

Worked Example: Building a Boxplot

Sorted dataset (N = 11): 6, 213, 241, 260, 281, 290, 314, 321, 350, 1500

(positions 1 through 10 above — note the original notes list 10 values but use N = 11 in the formula; worked as shown in the original notes below)

Median (Q2): location = $(50/100) \times 11 = 5.5 \rightarrow$ interpolating around position 5.5 gives Q2 = **285.5**

Q1: location = $(25/100) \times 11 = 2.75 \rightarrow 213 + 0.75 \times (241 - 213) = \mathbf{234}$

Q3: location = $(75/100) \times 11 = 8.25 \rightarrow 321 + 0.25 \times (350 - 321) = \mathbf{328.25}$

IQR = Q3 – Q1 = 328.25 – 234 ≈ **94**

Lower fence = Q1 – 1.5(IQR) = 234 – 1.5(94) ≈ **93**

Upper fence = Q3 + 1.5(IQR) = 328.25 + 1.5(94) ≈ **469**

Any values below 93 or above 469 are outliers. In this dataset, **6** (too low) and **1500** (too high) both fall outside the fences, so they are plotted as outlier points beyond the whiskers.

Benefits of a boxplot

- Easy way to see the distribution of data
- Shows the skewness of data
- Helps identify outliers
- Lets you compare two or more categories side by side

Side-by-side boxplots

When you have one numeric variable and one categorical variable (e.g., cat weight by sex), you can draw a boxplot for each category next to each other. This makes it easy to compare the distribution, median, and spread across groups — for instance, comparing the weight distributions of male vs. female cats side by side.

4. Scatterplots

A scatterplot shows the relationship between two numeric variables by plotting one on the x-axis and the other on the y-axis. Each point represents one observation. For example, plotting a home's ground living area (sq. ft.) against its sale price shows that larger homes tend to sell for more — a positive relationship — though the relationship isn't perfect (there's scatter, and a few outlier points).

Choosing the right plot for variable types

Variable Combination	Suggested Visualization
Continuous vs. Continuous	Scatterplot
Categorical vs. Numeric	Side-by-side boxplot
Categorical vs. Categorical	Cross-tabulation (crosstab)
Numeric vs. Numeric (grouped)	Scatterplot with color/hue by group

5. Covariance

What problem does covariance solve?

The plain average (mean) of a variable tells you its center, but not whether it moves together with another variable. Covariance answers: when one variable goes up, does the other tend to go up too (or down)?

What is covariance?

Covariance measures the degree to which two variables are linearly related — how much they change together.

Sign of Covariance	Meaning
Positive	Variables tend to move in the same direction
Negative	Variables tend to move in opposite directions
Zero (~0)	Variables are not linearly related

How is it calculated?

	Population	Sample
Formula	$\sigma_{xy} = \frac{\sum (X - \mu_x)(Y - \mu_y)}{n}$	$s_{xy} = \frac{\sum (x - \bar{x})(y - \bar{y})}{(n - 1)}$

X, Y = individual values | μ_x, μ_y = population means | $\bar{x}, \bar{y}$ = sample means | N, n = number of observations. The sample formula divides by (n–1) instead of n to correct for bias when estimating from a sample rather than a full population.

Worked Example: Experience vs. Salary

Experience (X)	Salary (Y)	X - $\bar{x}$	Y - $\bar{y}$	Product
2	1	-6	-5	30
5	2	-3	-4	12
8	5	0	-1	-1
12	12	4	6	24
13	10	5	4	20

$\bar{x} = 8$, $\bar{y} = 6$. Sum of products = 30+12–1+24+20 = 85.

Sample covariance = 85 / (5–1) = 85/4 = **21.25**

This positive covariance confirms: as experience increases, salary tends to increase too.

Contrasting Example: Backlogs vs. Package

Backlogs (X)	Package (Y)	X - $\bar{x}$	Y - $\bar{y}$	Product
--------------	-------------	---------------	---------------	---------

2	10	-6	4	-24
5	12	-3	6	-18
8	5	0	-1	0
12	2	4	-4	-16
13	1	5	-5	-25

$\bar{X} = 8$, $\bar{Y} = 6$. Sum of products = $-24-18+0-16-25 = -83$.

Sample covariance = $-83 / 4 = -20.75$

This negative covariance shows the opposite pattern: as backlogs increase, the salary package tends to decrease.

Limitation of covariance

Covariance tells you the *direction* of a relationship but not its *strength*. Its magnitude depends on the scale/units of the variables, so a covariance of 21 for one pair of variables isn't directly comparable to a covariance of 21 for a different pair measured in different units. This is the problem correlation solves.

6. Correlation

What problem does correlation solve?

Scatterplots can show relationships of varying strength — strong positive, weak positive, strong negative, weak negative, moderate negative, or no correlation at all. Correlation gives us a single number to **quantify** how strong and in what direction that relationship is, solving covariance's scale problem.

What is correlation?

Correlation measures the degree to which two variables are related and how they tend to change together, using the **correlation coefficient** (r), which always ranges from -1 to $+1$.

$$\text{Correlation} = \text{Cov}(x,y) / (\sigma_x \times \sigma_y)$$

Dividing covariance by the product of the two standard deviations rescales it into a standardized, unit-free number.

Value of r	Interpretation
$r = +1$	Perfect positive correlation
$r \approx 0.7 \text{ to } 1$	Strong positive correlation
$r \approx 0.3 \text{ to } 0.7$	Weak to moderate positive correlation
$r = 0$	No linear correlation
$r \approx -0.3 \text{ to } -0.7$	Weak to moderate negative correlation
$r \approx -0.7 \text{ to } -1$	Strong negative correlation
$r = -1$	Perfect negative correlation

7. Correlation vs. Causation

“Correlation does not imply causation” means that two variables being associated with each other does not necessarily mean one causes the other.

Classic Example: Firefighters and Fire Damage

There is a positive correlation between the number of firefighters present at a fire and the amount of damage caused. It would be wrong to conclude that firefighters *cause* more damage. A hidden third variable — the **severity of the fire** — explains both: more severe fires require more firefighters *and* cause more damage.

Correlations offer valuable insight into how variables relate, but establishing true causality requires stronger evidence — such as controlled experiments, randomized controlled trials, or carefully designed observational studies.

8. Visualizing Multiple Variables

When more than two variables are involved, a simple 2D scatterplot isn't enough. Here are common techniques for visualizing three or more variables at once.

Technique	What it shows
3D Scatter Plots	Plots three numeric variables at once on X, Y, and Z axes (e.g., separating iris species by petal/sep
Hue Parameter	Adds a categorical variable as color to a plot, e.g. a bar chart comparing duration by scenario, color
Facet Grids	Splits one plot into a grid of smaller plots, one per category (e.g., separate scatterplots for Lunch vs
Jointplots	Combines a central scatterplot with histograms of each variable along the margins, showing both th
Pairplots	A grid showing every pairwise scatterplot (and each variable's own histogram on the diagonal) acro
Bubble Plots	A scatterplot where a third numeric variable is encoded as the size of each point ('bubble'), and som

End of study guide — reorganized from Session 2 handwritten notes on Descriptive Statistics.